Bayesian model reduction selects supported structure I

The first two studies fixed the model's structure and asked how well its beliefs and learned parameters track the world; the third asks whether the model can also shed structure it never needed. The estimand is a Bayesian-model-reduction free-energy difference; the design is a categorical BMR diagnostic related to the structure-emergence mechanism discussed by Friston et al. (Friston et al. 2024), through the Bayesian-model-reduction lineage (Friston and Penny 2011). A full Dirichlet model carries a redundant state — one column the data never support — ranging over $n = 4$ candidate states. Bayesian model reduction scores swapping the prior for a *reduced* prior that prunes that column; the free-energy difference ΔF is the model-reduction objective eq. 16. This is a single deterministic evidence comparison, so there is no resampled sample and, by design, no confidence interval or paired test. The structure-learning frame here is the discrete-state model-selection thread the active-inference community has developed (Smith et al. 2022), applied to a colony's shared generative model.

ΔF is positive for the correct (redundant) pruning — the simpler model has more evidence and the run converges on it — and negative for the control pruning of a well-supported column, which is correctly rejected:

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- ΔF, pruning the **redundant** column: 3.68 (positive — reduction accepted)
- ΔF, pruning a **supported** column (control): -27.67 (negative — reduction rejected)
- Emergence converged (redundant accepted, supported rejected): Yes

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The sign pattern $\Delta F_{\text{redundant}} > 0 > \Delta F_{\text{supported}}$ is the demonstrated emergence verdict: the colony's generative model prunes the structure its data never support and retains the structure they do.

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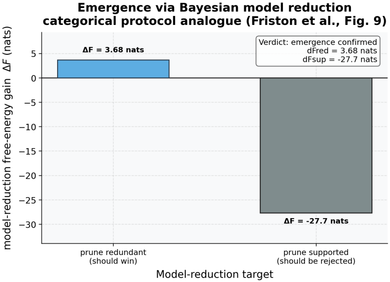

Figure 1: Bayesian-model-reduction (BMR) free-energy difference. Source relation: source-mechanism analogue to the model-reduction mechanism in Friston et al. (2024), Fig. 9; estimand: BMR ΔF in nats; uncertainty: deterministic closed-form comparison. ΔF for two candidate likelihood-column prunings in a colony with $n = 4$ hidden states. x-axis: the candidate pruning (redundant column vs. supported-column control); y-axis: ΔF in nats, where a positive value means the reduced model has more evidence and the pruning is accepted. The redundant-column bar is positive ($\Delta F = 3.68$ nats) — the data never supported this structure, so pruning it is the correct decision — while the supported-column control bar is negative ($\Delta F = -27.67$ nats), correctly rejected. The opposing signs constitute the emergence verdict (Yes). No error bar applies: BMR is a deterministic closed-form comparison on a single posterior.

fig. 1 contrasts the two prunings and annotates the convergence verdict.

Studies 1–3 all ran in a *trusting* world, where every broadcast belief is honest. The contamination sweep that follows removes that assumption, and it is the point at which the three robustness axes of sec. 7.2.1 begin to diverge.